DENIS SAMATOV

Senior Machine Learning Engineer — LLM/RAG & Agentic AI | Applied AI Technical Lead

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SUMMARY

Senior Machine Learning Engineer and Applied AI Technical Lead with 5+ years building production LLM/RAG and agentic AI systems across API models (GPT-4o, Claude Sonnet 4) and self-hosted open-weight and local-market models (DeepSeek, Gemma, GigaChat, YandexGPT/Alice) served via Ollama, orchestrated with LangChain and LangGraph using multi-provider model routing and tool calling. Delivered 3 production AI platforms end to end, leading 3-5-engineer teams, mentoring engineers, and partnering with product and research stakeholders while owning architecture, LLM evaluation (RAGAS, LLM-as-a-judge, custom golden-set harness), production readiness, and engineering quality. Works across Python, PyTorch, FastAPI, RAG and information retrieval, PostgreSQL, Qdrant, and Redis, with additional research depth in tensor methods, Bayesian inference, and radiomics.

SKILLS

AI / ML: hybrid retrieval, BM25, reranking, embeddings (OpenAI text-embedding-3, BGE, Sentence-Transformers, E5), full fine-tuning, instruction tuning, PyTorch, scikit-learn
LLM Models: GPT-4o, Claude Sonnet 4, DeepSeek V4 Flash / Pro, Gemma 4 31B, Gemma 3 12B, GigaChat, YandexGPT / Alice AI LLM
LLM Orchestration & Evaluation: LangChain, LangGraph, RAG, GraphRAG, Prompt Engineering, Tool Calling, MCP (Model Context Protocol), Multi-Agent Orchestration, Multi-Provider Model Routing, Inference Cost Optimization, Source-Grounded Generation, RAGAS, LLM-as-a-Judge
LLM Serving: Ollama, Ollama Cloud, self-hosted and API-based inference
Retrieval and Data: Qdrant, FAISS, PostgreSQL, SQL, Redis, Alembic, vector search, OCR, document ingestion, chunking, provenance, data lineage
Backend and MLOps: Python, FastAPI, REST APIs, Docker, RQ, MLflow, Airflow, Prometheus, GitHub Actions, CI/CD, automated testing, static analysis, observability, load testing, Git
Engineering Leadership: solution architecture, system design, ML/RAG design review, LLM evaluation strategy, agentic system design, mentoring, production readiness, engineering standards
Scientific and Medical ML: tensor decomposition, compressive sensing, sparse sensor placement, Bayesian inference, MCMC, medical image segmentation (U-Net, Attention U-Net, semantic segmentation), radiomics

PROFESSIONAL EXPERIENCE

Analytics and Machine Learning Department, MSUU

Machine Learning Engineer / Technical Lead | Apr 2024 - Present

- Delivered 3 production AI platforms - Professional Education AI Consultant, Knowledge Retrieval Platform, and Engineering Knowledge Copilot - from architecture through operational readiness, leading 3-5-engineer teams and owning technical design, delivery, and cross-system integration.
- Standardized OCR, hybrid retrieval, GraphRAG, reranking, provenance, and source-grounded generation into a single retrieval platform shared across all 3 AI products - orchestrated with LangChain and LangGraph - replacing independent per-product retrieval implementations.
- Established release-blocking LLM evaluation gates using RAGAS, LLM-as-a-judge, and a custom golden-set harness, achieving 18/18 golden QA cases at 5.0/5 — a 0% hallucination, false-escalation, and forbidden-output rate — through automated regression testing and CI quality gates.
- Extended platform capability with agentic workflows (tool calling, multi-agent orchestration via LangGraph) and multi-provider model routing across a mixed portfolio - GPT-4o and Claude Sonnet 4 via API, plus self-hosted open-weight and local-market models (DeepSeek, Gemma, GigaChat, YandexGPT/Alice) served via Ollama and Ollama Cloud - balancing accuracy, latency, and cost across production inference paths.
- Designed and implemented MCP-based integration layers across multiple AI systems, including a production-grade Yandex Workspace MCP server for Disk/Wiki workflows and additional MCP tools/connectors for agent access to external services, knowledge sources, and development environments, with typed schemas, access control, auditability, and stdio/HTTP transports.
- Productionized the /ask inference path to a validated operating envelope of 4.57 s p95 latency, 100% accepted-request success, 0 HTTP 5xx, and 0 timeouts, using controlled backpressure, concurrency limits, and failure-classified observability.
- Validated production-scale knowledge ingestion through 489 source records with zero registry failures and a 1,149-file live publication, using immutable manifests, content hashing, integrity verification, and rollback-safe staged delivery.
- Introduced architecture and ML/RAG design reviews, automated regression testing, static analysis, and CI release gates across 3-5-engineer teams, raising engineering consistency across all 3 platforms.

Heriot-Watt TPU Center

Data Scientist / Research ML Engineer (part-time) | Sep 2024 - Present

- Reduced full-field reconstruction error by 64.9%, from 0.57 to 0.20 relative Frobenius error, while achieving 0.88 SSIM and 37 dB PSNR from 10 instrumented wells, by developing tensor-based reduced-order models with QR-based sparse sensor placement and compressive-sensing recovery across 10 Brugge realizations.
- Extended sparse reconstruction to higher-accuracy operating points, reducing relative error by 80.7% versus the 1-well configuration and reaching 0.11 with 20 instrumented wells, by optimizing sensor placement for high-dimensional pressure and saturation fields.
- Accelerated a performance-critical NAB computation by 7.86x, verified through reproducible micro-benchmarks and a 538-test regression suite, by vectorizing cell-rank initialization while preserving numerical behavior.

Cardiology Research Institute

Machine Learning Engineer | Mar 2021 - May 2025

- Reduced end-to-end cardiac CT analysis time to 22.3 seconds per study on average by developing an automated pipeline covering preprocessing, epicardial adipose tissue segmentation, post-processing, and quantitative volume and density analysis.
- Delivered EPIFAT, a state-registered cardiac CT analysis system (Certificate No. 2025610317), integrating U-Net and Attention U-Net segmentation, anonymized preprocessing, manual ROI correction, radiomic analysis, and cross-platform application delivery.
- Transformed manual epicardial-fat quantification into an AI-assisted, human-in-the-loop workflow, automating segmentation and volume/density measurement while retaining operator ROI correction for controlled and reviewable outputs.

EDUCATION

Tomsk Polytechnic University

- Master's Degree, Applied Mathematics and Computer Science | 2024 - 2026
- Bachelor's Degree, Applied Mathematics and Computer Science | 2020 - 2024
- Diploma of Professional Retraining, Data Science and Machine Learning | 2023 - 2024

Skoltech (Skolkovo Institute of Science and Technology)

- Professional Development Course, Large Language Model-Based Agents | 2025
- Professional Development Course, Generative Models Based on Adversarial Learning | 2024

PUBLICATIONS AND RECOGNITION

Selected Publications

- "Tensor-Based Modal Decomposition and Sparse Sensor Placement for the Brugge Field Simulation Model." arXiv preprint, 2026.
- "Approach to Identifying Key Areas for Further Reservoir Study Using Tensor-Based Modal Decomposition." Conference paper, 2025.
- "Automatic Segmentation of Epicardial Fat and Quantitative Evaluation of Radiomic Parameters in Cardiac CT." Proceedings of the XXI International Conference "Perspectives of Fundamental Sciences Development," 2024.
- "Capabilities of Radiomic Analysis of Cardiac MRI Images in Cine Mode for Identifying Post-Infarction Areas." Digital Diagnostics.
- "Beam Parameters Restoration at the NICA Accelerator Complex Based on the Beam Position Monitor Data." Proceedings of START, Joint Institute for Nuclear Research, 2023.

Software Registration

- "EPIFAT - Module for Automatic Segmentation of Epicardial Adipose Tissue on Cardiac CT Images." Certificate of State Registration of Computer Program No. 2025610317, issued January 9, 2025.

Selected Awards, Conferences and International Programs

- Accepted Oral Presentation - "Tensor-Based Modal Decomposition with QR Pivoting for Sparse Sensor Placement and Field Reconstruction," AI4X - Accelerate Conference, Singapore, 2026.
- Accepted Presentation - "Probabilistic Evaluation of Parameter Space Using Neighbourhood Algorithm Bayes," Data Intelligence in the Oil and Gas Industry, Nizhny Novgorod, 2026.
- AI/ML Competition Awards - Winner, FINOdays AI/ML Track; Special Nomination Winner, National Technology Olympiad; Prize Winner, MIPT AI/Math/Physics Hackathon.
- Second-Degree Diploma - Cardiac CT segmentation and radiomics presentation, Perspectives of Fundamental Sciences Development, 2024.
- Participant, School-Conference on Tensor Methods in Mathematics and AI, Shenzhen, China, 2024.
- Participant, Skoltech-HIT Summer School, Harbin, China (remote), 2025.